

DSCI551 HW4

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1. `db.countrylanguage.countDocuments({IsOfficial: "T", Percentage: {$gt: 0.5}})`

```
> db.countrylanguage.countDocuments({IsOfficial: "T", Percentage: {$gt: 0.5}})
< 202
```

2. `db.country.find({Continent: /America/, GNP: {$gt: 100000}},`
`{_id: 0, Name: 1, GNP: 1})`

```
Name: 'Argentina',
GNP: 340238
}
{
  Name: 'Brazil',
  GNP: 776739
}
{
  Name: 'Canada',
  GNP: 598862
}
{
  Name: 'Colombia',
  GNP: 102896
}
{
  Name: 'Mexico',
  GNP: 414972
}
{
  Name: 'United States',
  GNP: 8510700
}
```

3. `db.country.aggregate([`

```
{ $match: { Continent: "North America", Population: { $gt: 100000 } } },  
{ $group: { _id: null, avgGNP: { $avg: "$GNP" } } } } )
```

```
> db.country.aggregate([  
  { $match: { Continent: "North America", Population: { $gt: 100000 } } },  
  { $group: { _id: null, avgGNP: { $avg: "$GNP" } } } ] )  
< {  
  _id: null,  
  avgGNP: 403444.625  
}
```

4. db.country.aggregate([

```
{ $lookup: { from: "city", localField: "Capital", foreignField: "_id.ID", as:  
"cap_city" } }, { $unwind: "$cap_city" }, { $sort: { Name: 1 } }, { $limit: 5 },  
{ $project: { _id: 0, countryName: "$Name", cityName: "$cap_city.Name" } } ] )
```

```
{  
  countryName: 'Afghanistan',  
  cityName: 'Kabul'  
}  
{  
  countryName: 'Albania',  
  cityName: 'Tirana'  
}  
{  
  countryName: 'Algeria',  
  cityName: 'Alger'  
}  
{  
  countryName: 'American Samoa',  
  cityName: 'Fagatogo'  
}  
{  
  countryName: 'Andorra',  
  cityName: 'Andorra la Vella'  
}
```

5. db.countrylanguage.aggregate([

```
{ $group: { _id: "$_id.Language", count: { $count: {} } } },
```

```
{ $match: { count: { $gt: 20 } }, { $sort: { count: -1, _id: 1 } } }
```

```
{
  _id: 'English',
  count: 60
}
{
  _id: 'Arabic',
  count: 33
}
{
  _id: 'Spanish',
  count: 28
}
{
  _id: 'French',
  count: 25
}
```

```
6. db.city.aggregate([
  { $match: { CountryCode: "USA" } },
  { $group: { _id: "$District", avgPop: { $avg: "$Population" }, count: { $count: {} } } },
  { $match: { count: { $gt: 10 } } }, { $project: { avgPop: 1 } } ])
```

```
{
  _id: 'Texas',
  avgPop: 354164.6538461539
}
{
  _id: 'California',
  avgPop: 245833.91176470587
}
{
  _id: 'Florida',
  avgPop: 210093.86666666667
}
```

```
7. db.city.insertMany([
```

```
{ "_id": { "ID": 4080 }, "Name": "LA suburb", "CountryCode": "USA", "District":  
"California", "Population": 1000},  
{ "_id": { "ID": 4081 }, "Name": "LA dessert", "CountryCode": "USA", "District":  
"California", "Population": 2000}}
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': {  
      ID: 4080  
    },  
    '1': {  
      ID: 4081  
    }  
  }  
}
```

```
8. db.city.updateOne(  
  { "_id.ID": 4081 },  
  { $set: { Name: "LA orange", Population: 3000 } })
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
9. db.city.deleteMany({ "_id.ID": { $in: [4080, 4081] } })
```

```
db.city.deleteMany({ "_id.ID": { $in: [4080, 4081] } })  
{  
  acknowledged: true,  
  deletedCount: 2  
}
```

Appendix (Results in MySQL):

This Appendix includes the results run in MySQL server for double-check.

count(Language)
202

Name	GNP
Argentina	340238.00
Brazil	776739.00
Canada	598862.00
Colombia	102896.00
Mexico	414972.00
United States	8510700.00

avg(GNP)
403444.625000

Name	Name
Afghanistan	Kabul
Albania	Tirana
Algeria	Alger
American Samoa	Fagatogo
Andorra	Andorra la Vella

Language	count(*)
English	60
Arabic	33
Spanish	28
French	25

District	avg(Population)
California	245833.9118
Texas	354164.6538
Florida	210093.8667